NATIONAL BIOINFORMATICS INFRASTRUCTURE SWEDEN

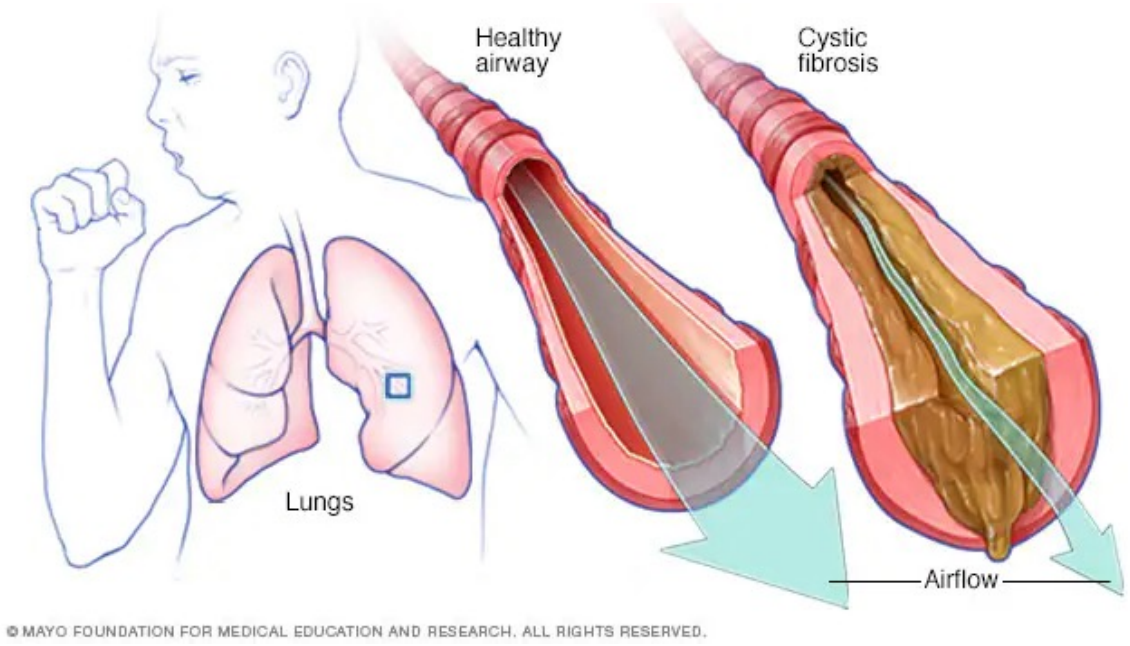
SciLifeLab

# Project – Main Assignment

# Practical information

- **Time**: Every afternoon, from ~15:00 to 17:00
- **Duration**: Throughout the entire week
- Apply your knowledge to tackle a real-world problem at a larger scale than the exercises
- Work on your own or in groups
- TAs are available for questions
- **Not mandatory** but highly recommended
- Solutions will be published on **Friday** after the lectures

# Background



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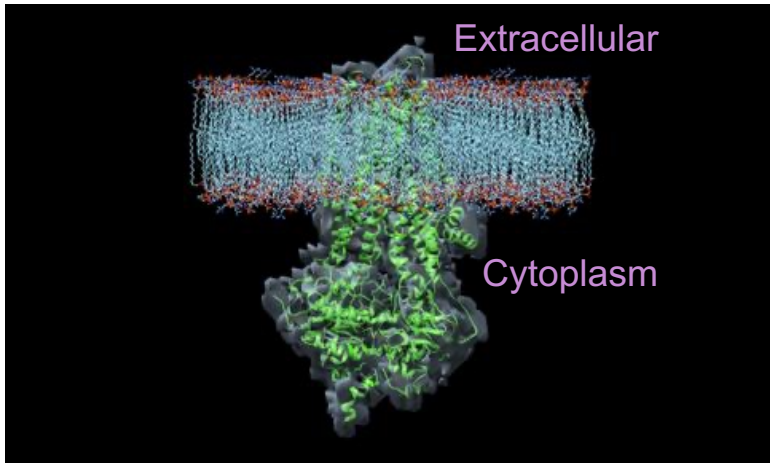
source: mayoclinic.org

## Cystic fibrosis (CF)

- Genetically inherited disease
- Produces thick and sticky mucus in organs, including the lungs and pancreas
- Clogs the airways and makes it difficult for patients to breathe
- No cure available
  - Previously only symptom management, such as airway clearance
  - Recent years developed modulator therapies. E.g. CFTR modulators, recover protein function for 90% of patients

# Genomic facts of Cystic Fibrosis

- CF is caused by mutations in Cystic Fibrosis Transmembrane Conductance Regulator (CFTR).
- The CFTR is an ion channel protein, acting like gates in a cell membrane that control the traffic of molecules through the membrane.
- For regular people, CFTR acts as a gate for chloride ions. When chloride leaves the cell, it carries water with it, which makes mucus less thick.
- For patients with CF, gene mutations in CFTR prevent this functionality, causing the mucus stays sticky and thick.




source: cff.org

# More about the CFTR gene

- The **CFTR gene** is located on **chromosome 7** in the human genome
- More than 1,500 mutations are known to cause cystic fibrosis (CF)
- One important class of mutations is non-synonymous mutations that introduce a **premature termination codon (PTC)**, resulting in a truncated (shortened) CFTR protein
- **We are particularly interested in mutations that create these premature stop signals**, because they prevent the CFTR protein from being fully produced

# Goal of the project

**Write Python program(s) that:**

- Extract the correct CFTR transcript from the human genome
- Translate it into its corresponding amino acid sequence
- Determine if one or more patients (from DNA sequences of 5 patients) have a premature stop codon, that is, **with shortened amino acid sequence**

**You will be guided step by step towards the final goal**

**For more details: Canvas -> Modules -> Project**

# Data

| Filename | Description |
| --- | --- |
| Homo_sapiens.GRCh38.93.gtf | Human reference annotation file (GTF format) |
| Homo_sapiens.GRCh38.dna_sm.chromosome.7.fa | Human reference DNA sequence for chromosome 7 (fasta format) |
| Patient1.fa | DNA sequence data of chr 7 for Patient 1 (fasta format) |
| Patient2.fa | DNA sequence data of chr 7 for Patient 2 (fasta format) |
| Patient3.fa | DNA sequence data of chr 7 for Patient 3 (fasta format) |
| Patient4.fa | DNA sequence data of chr 7 for Patient 4 (fasta format) |
| Patient5.fa | DNA sequence data of chr 7 for Patient 5 (fasta format) |

# Fasta format

```
>MT dna:chromosome chromosome:GRCh38:MT:1:16569:1 REF
GATCACAGGTCTATCACCCTATTAACCACTCACGGGAGCTCTCCATGCATTTGGTATTTT
CGTCTGGGGGGTATGCACGCGATAGCATTGCGAGACGCTGGAGCCGGAGCACCCTATGTC
GCAGTATCTGTCTTTGATTCCTGCCTCATCCTATTATTTATCGCACCTACGTTCAATATT
ACAGGCGAACATACTTACTAAAGTGTGTTAATTAATTAATGCTTGTAGGACATAATAATA
ACAATTGAATGTCTGCACAGCCACTTTCCACACAGACATCATAACAAAAAATTTCCACCA
AACCCCCCCTCCCCCGCTTCTGGCCACAGCACTTAAACACATCTCTGCCAAACCCCAAAA
ACAAAGAACCCTAACACCAGCCTAACCAGATTTCAAATTTTATCTTTTGGCGGTATGCAC
TTTTAACAGTCACCCCCCAACTAACACATTATTTTCCCCTCCCACTCCCATACTACTAAT
CTCATCAATACAACCCCCGCCCATCCTACCCAGCACACACACACCGCTGCTAACCCCATA
CCCCGAACCAACCAAACCCCAAAGACACCCCCCACAGTTTATGTAGCTTACCTCCTCAAA
```

# GTF format

- GTF stands for Gene transfer format
- Holds information about the gene structure
- Tab-delimited, i.e., columns are separated by tabs (‘\t’)

# GTF example

```
#!genome-build GRCh38.p12
#!genome-version GRCh38
#!genome-date 2013-12
#!genome-build-accession NCBI:GCA_000001405.27
#!genebuild-last-updated 2018-01
1 havana gene 11869 14409 . + . gene_id "ENSG00000223972"; gene_version
  "5"; gene_name "DDX11L1"; gene_source "havana"; gene_biotype
  "transcribed_unprocessed_pseudogene";
1 havana transcript 11869 14409 . + . gene_id "ENSG00000223972";
  gene_version "5"; transcript_id "ENST00000456328"; transcript_version
  "2"; gene_name "DDX11L1"; gene_source "havana"; gene_biotype
  "transcribed_unprocessed_pseudogene"; transcript_name "DDX11L1-202";
  transcript_source "havana"; transcript_biotype
  "processed_transcript"; tag "basic"; transcript_support_level "1";
```

# GTF example

```
<seqname> <source> <feature> <start> <end> <score> <strand> <frame> [attributes]

1 havana gene 11869 14409 . + . gene_id "ENSG00000223972"; gene_version "5"; gene_name "DDX11L1"; gene_source "havana"; gene_biotype "transcribed_unprocessed_pseudogene";
```

# Columns of GTF file

**<seqname> <source> <feature> <start> <end> <score> <strand> <frame> [attributes]**

1. **<seqname>**: The name of the sequence (typically a chromosome number).
2. **<source>**: The source of the annotation (e.g., ENSEMBL).
3. **<feature>**: The type of feature (e.g., gene, transcript, exon).
4. **<start>**: The starting position of the feature in the sequence, **starting from 1**.
5. **<end>**: The ending position of the feature in the sequence, **end position included**.
6. **<score>**: A score between 0 and 1000, or . if not applicable (indicating the reliability of the annotation).
7. **<strand>**: The strand on which the feature is located (`+` for the forward strand, `-` for the reverse strand). (**only forward strand in this project**)
8. **<frame>**: The reading frame, one of '0', '1' or '2', or `.` if not applicable.
9. **[attribute]**: A list of key-value pairs providing additional information about the feature.

# Attribute of GTF

- **A semicolon (;)**-separated list of key-value pairs
- For each key-value pair, key is one word, and the value is quoted by double quotes, which may contain multiple words
- A key can be repeated multiple times.

**Some attributes (always semi-colon separated key-value pairs):**

- gene_id: The stable identifier for the gene
- gene_version: The stable identifier version for the gene
- gene_name: The official symbol of this gene
- gene_source: The annotation source for this gene
- transcript_id: The stable identifier for this transcript
- transcript_name: The symbold for this transcript derived from the gene name
- exon_id: The stable identifier for this exon

# How to get DNA sequence of a gene

```
<seqname> <source> <feature> <start> <end> <score> <strand> <frame> [attributes]

1 havana gene 11869 14409 . + . gene_id "ENSG00000223972"; gene_version "5"; gene_name "DDX11L1";
gene_source "havana"; gene_biotype "transcribed_unprocessed_pseudogene";
```

- Get full DNA sequence of human ch7 from the file "Homo_sapiens.GRCh38.dna_sm.chromosome.7.fa"
- The position is 1-based, with both ends included

```
start_pos = 11869
end_pos = 14409

seq_gene = ch7_dna_sequence[start_pos - 1:end_pos]
```

# Getting started

- Create a folder called project
- Download and extract the project files in this folder (download link is provided in the project page)
- Work with Jupyter or any text editor or IDE (e.g., Spyder, VSCode)
- Ask questions if something is unclear
- Speak to a TA or discuss with your neighbour
- Or use the discussion section in Canvas
- Don’t be too fast ☺
  - Day 1: >| question 2
  - Day 2: >| question 5
  - Day 3: The whole project
- Find out more info at Canvas -> Modules -> Project

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# Questions?